Mark Julius P. Mella

Software Engineer | Agentic AI Engineer

Email: markmella.dev@gmail.com
Phone: 9625881552
GitHub: github.com/mark-dev0x

Professional Summary

Software engineer with 6+ years across freelance and product development, transitioning into agentic AI engineering. Plan and build web applications with full-stack knowledge, REST APIs and persistent data. Strong in frontend development, performance, caching, SEO and output validation.

Technical Skills

Programming and frontend: JavaScript, TypeScript, Vue 2 / Vue 3, React, Next.js, HTML, CSS, Sass, Tailwind CSS

APIs and state management: Vue Router, Pinia, Vuex, REST APIs, Axios, TanStack Vue Query, SignalR, Zustand, Zod, React Hook Form, Mobile WebViews

Performance and SEO: Caching, IndexedDB, Data fetching, Image optimization, Lazy loading, Code splitting, Hydration, Lighthouse, Search engine optimization (SEO)

Agentic AI workflows: Cursor, Claude Code, Codex, ChatGPT, AI-assisted research, Structured prompts, Task decomposition, Context and constraints, Output validation, Automation knowledge

Backend, data and development tools: Node.js, PHP, Laravel, Firebase, Firestore, Chart.js, Vite, Git, Figma-to-code, Agile / Scrum

Work Experience

Software Developer — Frontend & API Integration

May 2022 - Present

DigiPlus Interactive Corp

- Integrate REST APIs and real-time features for sports, account management, deposits, withdrawals, and customer support.
- Develop and maintain sportsbook and SEO-focused web applications using Vue 3, TypeScript, Vite, and Tailwind CSS.
- Build responsive user interfaces, reusable components, shared layouts, design tokens, and light and dark themes for web and mobile.
- Optimize application loading through caching, image and font delivery, lazy loading, and code splitting.
- Support Vue 2-to-Vue 3 migrations and collaborate with product managers, designers, and backend engineers.

Frontend Developer

May 2022 - May 2023

Solid Leisure Solutions

- Maintained production web applications and delivered frontend updates.
- Translated Figma designs into responsive user interfaces.
- Integrated REST APIs to connect interfaces with backend services.
- Integrated customer-support chat into web applications.
- Optimized frontend performance to improve application responsiveness.

Education

Bachelor of Science in Computer Science

2018 - 2022

Pamantasan ng Lungsod ng Pasig

Projects

Frontend Performance, Themes and Retail Interfaces

Integrated REST APIs for account and wallet journeys. Built shared themes and authenticated layouts; optimized image delivery, SDK loading and caching.

Vue 3, TypeScript, REST APIs, Pinia, Vite

Product Storefront and Inventory Management

Contributed image loading, description fallbacks and version-aware caching. Added subcategory management, Firestore access rules, publication controls and cache refresh.

React, Next.js, TypeScript, Firebase, IndexedDB

Catalog Editing Workspace

Contributed multi-catalog management, saved design settings and database-backed item ordering with optimistic updates and rollback.

React, Next.js, TypeScript, Zustand, Zod, Firestore

Campaign Forms, Event Draws and Reporting

Adjusted receipt upload and participation states; updated draw participant loading and animation. Added duplicate receipt review, date filters and reporting views.

React, Next.js, Vue 3, TypeScript, Firebase, Chart.js

Agentic AI Workflows and Automation

My foundations began with self-study in college in 2018. I started learning AI in early March 2025 and am now transitioning into agentic AI engineering.

- Use Codex and ChatGPT for research, and Cursor and Claude Code for code audits, debugging, and incremental refactoring.
- Structure prompts with context, constraints, and reviewable tasks; validate output through code review, type checks, builds, and browser inspection.
- Apply knowledge of automation to task planning while reviewing and validating AI-generated code.

Additional Work Experience

Frontend Developer - Internship

March - April 2022

Public Safety Savings and Loan Association, Inc. (PSSLAI)

- Built employee risk-calculation interfaces for a risk management application using React, Node.js, and AWS.

Freelance Frontend Developer

2020 - 2022

- Translated Figma designs into responsive web and mobile layouts, and built reusable UI components.

Freelance Computer Technician

2020 - 2022

- Installed and configured hardware and software; diagnosed and repaired desktop and laptop issues.

Awards and Certifications

- Best in Thesis - AWIT, an interpreted Turing-complete scripting language written in C with bytecode generation and garbage collection.
- Dean's Lister, 2021-2022.
- Google Tech Challenge certificate, September 28, 2019.